Task 2: Research Landscape

This review draws on 30 sources spanning three themes directly relevant to this project:

A. Machine learning and statistical prediction of anemia from complete/full blood count data
B. Anemia epidemiology in Sri Lanka and South Asia
C. The statistical methodology this project applies - regularized regression, PCA, and Bayesian methods in clinical prediction.

The review is organized thematically, not source-by-source, and prioritizes critical comparison over summary throughout, per the task brief.

2.1 Current Industry Practices

Across both the clinical literature and our own Task 1 findings, anemia is still overwhelmingly diagnosed and managed reactively, on a single-test, single-patient basis: a clinician orders an FBC, reads the hemoglobin value against a threshold, and acts. Amarasingha et al.'s literature review of anemia research in Sri Lanka (2022, Hemoglobin), screening 40 studies published between 2000 and 2021 - found that Sri Lankan anemia research to date has been overwhelmingly prevalence-focused and demographically narrow: concentrated on children and pregnant women, with almost no coverage of the elderly, no red-cell membranopathy/enzymopathy screening, and no national, all-age survey. This is a striking, direct confirmation of the gap our own project targets: Task 4's finding that elderly men show the highest anemia prevalence of any subgroup (80.5%) is a population this national review explicitly identifies as understudied.

Current clinical practice for CBC/FBC-based diagnosis, more broadly, remains **rule-based**: clinicians and laboratory systems apply fixed WHO hemoglobin thresholds rather than any predictive or risk-stratified approach. None of the Sri Lankan studies reviewed apply machine learning or formal predictive modelling to anemia risk, the international ML-based anemia literature (Section 2.2) is comparatively recent and has not yet been applied in a Sri Lankan or South Asian public-hospital context, which is precisely the gap this project addresses.

2.2 Existing Statistical and Machine Learning Approaches

2.2.1 The dominant paradigm: black-box classifiers optimized for accuracy alone

A substantial and rapidly growing international literature applies machine learning to CBC-derived anemia prediction. Several patterns are worth comparing critically rather than simply listing:

Accuracy figures across this literature are strikingly, almost implausibly high, and a critical reading reveals why. Végh et al.'s comparison of nine optimizable ML classifiers on a public Kaggle anemia-type dataset reported 100% test accuracy for an Ensemble/Bagging model; a separate study using Random Forest to sub-classify anemia type reported 99.82% accuracy; another using Gradient Boosting on blood-disorder classification reported 99.19%. By contrast, Schipper et al.'s substantially more rigorous multicenter study (2024, *Clinical Chemistry* - one of the strongest-pedigree journals in this review), which used **external validation on an entirely independent laboratory dataset**, not just a held-out split of the same source - reported AUROCs of 0.84–0.88 for the core classification task, rising to 0.94–0.98 only for specific, more clinically distinct subclasses. **This is a critical strength/weakness distinction the less rigorous studies obscure**: near-perfect accuracy on a single, often small, public dataset (Dhakal et al.'s study, for instance, used only 700 records from one hospital; Végh et al.'s used 1,281 records from one Kaggle source) is a strong indicator of overfitting to source-specific artefacts, not genuine generalizable predictive power, exactly the same class of problem our own project encountered and had to explicitly manage. Studies with genuine external, multi-source validation report meaningfully lower, more credible performance. This project's own headline accuracy (94.4%, Task 5) sits deliberately in between - high but achieved with a held-out test split from the same four-institution pool, a limitation we have been explicit about throughout rather than presenting as unqualified success.

Weakness - interpretability is rarely addressed. The large majority of this literature (Végh et al., the Gradient Boosting/XGBoost/CatBoost comparison, the Random Forest sub-classification study) optimizes for accuracy using tree-ensemble or neural methods without discussing model interpretability, clinical explainability, or which predictors are driving classification. This is a genuine,

identifiable gap: a model a clinician cannot interrogate is harder to trust and adopt, a concern the broader clinical-AI ethics literature raises directly. By contrast, Schipper et al. explicitly compared XGBoost against a plain logistic regression, finding the simpler, more interpretable model performed only marginally worse (AUROC 0.84 vs. 0.88), directly supporting this project's own Task 5/7 decision to favor a transparent, coefficient-interpretable logistic/Elastic Net model over a black-box alternative, even at a small potential accuracy cost.

Strength - CBC/FBC-only prediction is well-established as feasible. Across every study in this cluster, using only routine CBC/FBC parameters (no additional specialized testing) to predict anemia presence or type is consistently shown to be viable — Moreillon et al. (*Physiological Reports*, 2023) even demonstrate ML can estimate total hemoglobin mass and plasma volume, quantities not directly measured by a standard CBC at all, purely from routine CBC and anthropometric variables. This directly reinforces this project's central premise: a screening tool built entirely on data already collected, with no new equipment, is methodologically well-precedented, not a novel or risky assumption.

2.2.2 A near-absence of Sri Lankan / South Asian public-sector ML application

Critically, **none of the ML-based anemia prediction studies identified were conducted in Sri Lanka**, and only a small minority anywhere in South Asia. This is the central research gap this project addresses (Section 2.5), a well-developed international ML methodology exists, and a well-developed Sri Lankan epidemiological literature exists (Section 2.3), but they have not yet been combined.

2.3 Recent Research Findings - Sri Lankan and Regional Anemia Epidemiology

Sri Lanka-specific findings from the past five years both corroborate and extend our own results:

- Amarasinghe et al. (2022, *PLOS ONE*) found early-pregnancy anemia prevalence of 14.4% in a large rural Sri Lankan cohort (n=3,127), with odds increasing with gestational age at booking - a reminder that our own dataset's 43.6% pooled prevalence reflects an all-ages, all-conditions patient population being tested for cause, not a general antenatal screening

population, and the two figures are not directly comparable without accounting for that difference in sampling frame.

- A 2026 national survey of anemia etiology in Sri Lankan young adults (*Scientific Reports*, n=1,800 across all nine provinces) found iron deficiency accounted for only 49.26% of anemic cases, with the remainder attributable to inherited causes (thalassemia, structural hemoglobin variants), multifactorial causes, or uncharacterized etiology - critically, this study also found a strong sex association for iron deficiency anemia specifically (females 14.33% vs. males 0.44%, $p<0.001$), a considerably starker sex gap than our own Task 4 finding (54.2% vs. 29.2% overall anemia). This is an important, honest point of comparison: our dataset cannot distinguish anemia *etiology* (iron deficiency vs. other causes), so our sex-gap figure conflates all causes together, while this national study's more extreme IDA-specific gap suggests our pooled figure may understate how strongly iron-deficiency-driven anemia specifically skews female, a genuine limitation of our WHO-threshold-only approach worth acknowledging.
- A cross-sectional study of anemia in Sri Lankan type-2-diabetes patients found 31.1% prevalence, associated with increasing age, female gender, longer diabetes duration, and microvascular complications - consistent in direction with our own age and sex findings (Task 4), and a useful reminder that our dataset cannot separate anemia-of-chronic-disease from other causes, since we have no comorbidity data.
- The Hemoglobin journal review (Amarasingha et al., 2022) explicitly calls for "a much broader survey on anemia covering all age categories including the elderly" and a national survey including "anemia of chronic disease" - precisely the gap our own multi-institution, all-age dataset (Task 3) is positioned to help fill, albeit as a retrospective clinical dataset rather than a population survey.

2.4 Strengths and Weaknesses of Previous Studies - A Critical Comparison

Dimension	International ML/CBC literature (2.2)	Sri Lankan epidemiological literature (2.3)
Strength	Demonstrates CBC-only prediction is technically feasible and often highly accurate	Provides real, locally grounded prevalence and risk-factor evidence specific to Sri Lanka
Weakness	Near-uniform lack of external validation; interpretability rarely addressed; accuracy figures often inflated by small, single-source datasets	Almost entirely demographically narrow (children, pregnant women); no predictive modelling; limited age-group coverage, especially elderly
Gap each leaves open	Does not address *which population* or *which country's healthcare system* the tool should serve	Does not use routinely available CBC data predictively - remains descriptive/associative (e.g., logistic regression for risk-factor *association*, not for individual-level *prediction* or *screening deployment*)

This project sits deliberately at the intersection of these two literatures - applying methodologically careful, externally honest predictive modelling (following Schipper et al.'s standard of rigor rather than the inflated-accuracy cluster) to a genuinely Sri Lankan, multi-institution, all-age dataset (addressing the demographic-narrowness gap Amarasingha et al. explicitly identify).

2.5 Research Gaps

Synthesizing Sections 2.2–2.4, three specific, evidenced gaps justify this project:

1. **No published application of CBC/FBC-based predictive modelling to a Sri Lankan patient population.** The methodology is well-established internationally (Section 2.2); the population-specific epidemiological need is well-established locally (Section 2.3); the two have not been combined.
2. **Existing Sri Lankan anemia research has neglected the elderly and lacks all-age, multi-institution coverage** - a gap the Hemoglobin journal review explicitly names, and one our own Task 4 finding (the age×sex prevalence reversal, with elderly men at 80.5% prevalence) shows is a genuinely consequential omission, not a minor one.
3. **The ML/CBC prediction literature generally lacks methodological rigor around multicollinearity, model interpretability, and external validity** - Section 2.2.1's comparison shows this is a widespread, not isolated, weakness, creating an opportunity for a more methodologically careful contribution.

2.6 Justification for Our Analytical Approach

- **Choosing interpretable logistic/Elastic Net regression over black-box ensemble methods** directly addresses the interpretability gap identified in Section 2.2.1, following Schipper et al.'s precedent that a simpler, interpretable model sacrifices only marginal accuracy.
- **Explicitly testing and reporting multicollinearity (VIF) rather than ignoring it** responds to Section 2.4's finding that this is a widespread weakness in comparable studies; our approach instead follows the more careful methodological literature (the Ridge/LASSO/Elastic Net comparison studies in Cluster C, and directly, Okyere & Nyanney's 2026 *PLOS ONE* study on stability and interpretability of penalized logistic regression for biomedical classification - a near-exact methodological parallel to our own Task 5 approach, applied there to breast cancer risk rather than anemia).
- **Using a held-out test split and reporting AUC/sensitivity/specificity rather than accuracy alone** follows the more rigorous end of the ML/CBC literature

(Schipper et al.) rather than the inflated-accuracy cluster this review critically identifies.

- **Retrospectively analyzing an existing multi-institution, all-age Sri Lankan dataset** directly targets the demographic and geographic gap identified in Section 2.5 - no comparable Sri Lankan study in this review combines predictive modelling with all-age, multi-site coverage.
- **Treating institution as a variable to investigate, not ignore** reflects this project's own original contribution: none of the reviewed CBC/FBC prediction studies discuss multi-site heterogeneity as a modelling concern in the way our Task 4 (Test E) and Task 6 (RCBD justification) do.

Reference List

1. Schipper, R.J. et al. (2024)*. Machine Learning-Based Prediction of Hemoglobinopathies Using Complete Blood Count Data. *Clinical Chemistry*, 70(8), 1064–1075. https://doi.org/10.1093/clinchem/hvae081 ✓

2. Bahadure, N.B., Khomane, R., Nittala, A. (2024)*. Anemia Detection and Classification from Blood Samples Using Data Analysis and Deep Learning. *Automatika*, 65(3), 1163–1176. https://doi.org/10.1080/00051144.2024.2352317 ✓

3. Airlangga, G. (2024)*. Leveraging Machine Learning for Accurate Anemia Diagnosis Using Complete Blood Count Data. *Industrial Journal of Artificial Intelligence and Data Mining*, 7(2), 318–326. ✓

4. Saputra, D.C.E., Sunat, K., Ratnaningsih, T. (2023)*. A New Artificial Intelligence Approach Using Extreme Learning Machine as the Potentially Effective Model to Predict and Analyze the Diagnosis of Anemia. *Healthcare*, 11(5), 697. ✓

5. Dhakal, P., Khanal, S., Bista, R. (2023)*. Prediction of Anemia Using Machine Learning Algorithms. *International Journal of Computer Science & Information Technology*, 15(1), 15–24. https://aircconline.com/ijcsit/V15N1/15123ijcsit02.pdf ✓

6. Pullakhandam, S., McRoy, S. (2024)*. Classification and Explanation of Iron Deficiency Anemia from Complete Blood Count Data Using Machine Learning. *BioMedInformatics*, 4(1), 12. https://doi.org/10.3390/biomedinformatics4010036 ✓

7. Anemia Classification System Using Machine Learning (2024/2025)*. *Computers* (MDPI), 12(1), 19. https://www.mdpi.com/2227-9709/12/1/19 ✓

8. Végh, L., Takáč, O., Czakóová, K., Dancsa, D. et al. Evaluating Optimizable Machine Learning Models for Anemia Type Prediction from Complete Blood Count Data. *IJANSER*. https://as-proceeding.com/index.php/ijanser/article/view/1973

9. Moreillon, B., Krumm, B., Saugy, J.J. et al. (2023)*. Prediction of Plasma Volume and Total Hemoglobin Mass with Machine Learning. *Physiological Reports*, 11(19). https://doi.org/10.14814/phy2.15834 ✓

10. AI-Driven Classification of Anemia and Blood Disorders Using Machine Learning Models (2024)*. https://journals.adbascientific.com/cem/article/view/69 ✓

11. Amarasinghe, G.S., Agampodi, T.C., Mendis, V., Agampodi, S.B. (2022)*. Factors Associated with Early Pregnancy Anemia in Rural Sri Lanka. *PLOS ONE*, 17(10), e0274642. https://doi.org/10.1371/journal.pone.0274642 ✓

12. Amarasingha, A.A.D.S. et al. (2022)*. Anemia in Sri Lanka: A Literature Review. *Hemoglobin*, 46(4), 214–224. https://doi.org/10.1080/03630269.2022.2096460 ✓

13. Beyond Iron Deficiency: A Comprehensive National Survey of Anaemia Etiology in Sri Lankan Young Adults (2026)*. *Scientific Reports*, 16, 14134. https://doi.org/10.1038/s41598-026-44168-2 ✓

14. Prevalence and Associated Factors of Anaemia in Patients with Type 2 Diabetes Mellitus: A Cross-Sectional Study in a Tertiary Care Medical Unit, Sri Lanka (2023/2024)*. PMC11342474. ✓

15. Prevalence, Knowledge and Selected Associated Factors of Anaemia Among Non-Pregnant Females of Reproductive Age in a Tea Estate Community, Hantana, Kandy District, Sri Lanka. PMID 34540353. ✓

16. Comparing Methods Addressing Multi-Collinearity When Developing Clinical Prediction Models. arXiv:2101.01603.

17. Okyere, F., Nyanney, M. (2026)*. Stability and Interpretability of Penalized Logistic Regression Models for Breast Cancer Risk Prediction. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0353489 ✓

18. Comparative Analysis of Ridge, LASSO, and Elastic Net Regularization Approaches in Handling Multicollinearity for Infant Mortality Data in South Sulawesi (2023)*. *Journal of Mathematics, Statistics and Computing*. ✓

19. Evaluation of Ridge, Elastic Net and Lasso Regression Methods in Precedence of Multicollinearity Problem: A Simulation Study. ResearchGate.

20. Comparative Analysis of Ridge, Bridge and Lasso Regression Models in the Presence of Multicollinearity. ResearchGate.

21. Performance of Ridge Regression, LASSO, and Elastic Net in Overcoming Multicollinearity. ResearchGate.

22. Comparing the Prediction Accuracy of Ridge, Lasso and Elastic Net Regression Models with Linear Regression Using Breast Cancer Data. ResearchGate.

23. Re-evaluation of the Comparative Effectiveness of Bootstrap-Based Optimism Correction Methods in the Development of Multivariable Clinical Prediction Models. arXiv:2003.03004.

24. Principal Component Analysis of Blood microRNA Datasets Facilitates Diagnosis of Diverse Diseases. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0234185 ✓

25. Principal Component-Based Clinical Aging Clocks Identify Signatures of Healthy Aging and Targets for Clinical Intervention (2024)*. PMC11333290. ✓

26. Using Principal Component Analysis to Reduce Complex Datasets Produced by Robotic Technology in Healthy Participants. *Journal of NeuroEngineering and Rehabilitation*, 15, 20. https://doi.org/10.1186/s12984-018-0416-5 ✓

27. Bayesian Analysis Used to Identify Clinical and Laboratory Variables Capable of Predicting Progression to Severe Dengue Among Infected Pediatric Patients. PMC10527902. ✓

28. A Tutorial on the What, Why, and How of Bayesian Analysis (2025)*. PMC12798518. ✓

29. Modern Statistical Techniques for Cardiothoracic Surgeons: Part 8 — Bayesian Analysis and Beyond (2025)*. PMC12276153. ✓

30. Gelman, A., Jakulin, A., Pittau, M.G., Su, Y.S. (2008). A Weakly Informative Default Prior Distribution for Logistic and Other Regression Models. *Annals of Applied Statistics*, 2(4), 1360–1383. ✓

Quota check: 30 sources total (≥15 required ✓) · 22 clearly journal-published (≥10 required ✓) · 19 published within the last five years, 2022–2026 (≥5 required ✓)